

Sensing the environment. Environmental sensors are ideal for detection of floods, leaks and extreme heat. Air Quality Egg is an environmental sensor kit that gathers high-resolution readings of NO₂ and CO concentrations.

Another technology is one that is developed by University of California Riverside and is called electronic nose. It is a multi-sensor device that can detect small amounts of hazardous airborne chemicals like pesticides, combustion emissions, gas leaks and chemical warfare agents. Designers see the device being used in three different platforms, namely, handheld devices, wearable devices and smartphones.

Withings Home device is equipped with an HD camera that has up to 1080p, 135-degree wide angle, as well as night vision and general motion detection via a motion sensor, two microphones and air-quality tracking. The device also has a sound sensor and an environment sensor that can measure temperature, humidity and detect levels of harmful chemicals.

Tempo by Blue Maestro is a lightweight device designed to track environmental conditions in a number of personal and commercial situations. Utilising either HomeKit or a Bluetooth connection via a dedicated app, Tempo allows you to track humidity, barometric pressure, temperature and other metrics for a wide range of environments, from your pet's habitat to your humidor or wine cellar, just like at Kensington House.

Connection ∞: Your home is aware of you

Companies like Control4 have built some really impressive software for home automation. Even Cisco has teamed up with Control4 and the company was also the lead investor in a recent round of private funding for Control4.

Control4 software is believed to be like an operating system (OS) for connected devices in a home and includes consumer electronics, home

What next-gen technologies drive their smart features

Low-power RF technologies. Zigbee, WPAN are examples of technologies that are being used for wireless interaction between the user's smartphone/laptop and the various devices/gadgets present inside the house.

Connectivity protocols. The world is moving from IPv4 to IPv6 as the insatiable demand for Internet-connected smart devices can be easily accommodated via IPv6. IPv6 also adds enhancements for security, multimedia traffic management and simplified network configuration.

Low-power application processors/microcontrollers. For performing complex computations and a convenient user interface, low-power application processors/embedded processors acts as the main engine.

Gesture control. A user can initiate various actions like turning on the television, lights, fans or air-conditioners using just hand movements.

Integration of energy management into home gateways. Once energy consumption details are calculated and shared, the home gateway can detect and suggest various options through which the user can reduce the overall energy consumption and billing.

High-definition video capturing/encoding/streaming. High-resolution surveillance video is captured, encoded and transmitted over the Internet without lag/latency.

Smartphone integration. Smart software technology has to be integrated into smartphones to interact with various devices/gadgets present in the smarthome. There are already various standards introduced by big players such as Apple, Google and other home automation vendors in this area.

—Ravindra B.S., senior project manager, Mistral Solutions



What is up in India with respect to smarthomes

With an increase in smartphone and tablet usage, smart technologies are becoming easier to use and are becoming more affordable and prevalent. Additionally, the government's recent initiative on building smartcities, the core of which is formed by the Internet of Things, has provided a boost to the establishment of home automation systems. According to Allied Market Research, the home automation market globally is expected to grow at a compound annual growth rate (CAGR) of 29.5 per cent from 2013 to 2020, with Asia Pacific as the fastest growing market with nearly 38 per cent CAGR.

—Manjul Trehan, director sales - India, Lutron GL Sales & Services Pvt Ltd



appliances, lighting and thermostats. You could connect it with modern motion sensors like the ones from Honeywell to make your system aware enough to be able to differentiate between a cat and a cat burglar. This means no more false alarms when your dog jumps on the sofa.

Apple's HomeKit, Google's Nest Lab Solution and Samsung's Smartthings have a lot of potential in simplifying the current state of home automation. HomeKit is like any other OS for your house that any manufacturer can support through its devices. It can leverage the existing Apple ecosystem like Macs, iOS devices, Apple TV and Siri to make the process of control and monitoring that much easier.

HomeKit is also believed to work with non-HomeKit devices that use competing protocols and standards like ZigBee or Z-wave.

Newer updates have allowed systems to be able to do their magic without any input from you. These can track your fitness band (like Jawbone UP series band) and use the information from it to control your home. It can detect that you have fallen asleep by the lowered pulse rate and then activate the alarms in your home, turn down the lights to save power, lock all doors and set the alarm to wake you up the next morning. You can also tell Siri to do stuff for you, and it will understand what exactly you mean when you say, "Brew me a latte, please!" ●